

PV239 – 05 Storage

Project status

- Teams
- Topic submission
- Discord
- Consultations

Goals

- Storing values
- Storing values securely
- Storing data locally
- Sending messages between different application parts

Preferences

- For storing small amounts of values
- Limit for size of individual records
- Synchronous API
- Supported types:
 - bool
 - int, long
 - float, double
 - string
 - DateTime

Preferences

- Android
 - Stored in Shared Preferences
 - Enables automatic backup to Google Drive – from Android 6.0 (API level 23)
- iOS
 - Stored in UserDefaults
 - Backup in iCloud

Preferences

DEMO

Secure Storage

- Useful for storing small amounts of data securely (i.e. authentication token)
- Supports only strings
- Asynchronous API

Secure Storage

- Android
 - Stored in Shared Preferences
 - Encrypted using a key from Android KeyStore
- iOS
 - Stored in KeyChain
 - Backup is on/off based on user's settings

Secure Storage

DEMO

SQLite

- Nuget package sqlite-net-pcl from author „Frank A. Krueger“
- We need to set a path to the database file
- Basic ORM
- Basic operations

Table	Query
Insert	Execute
Update	ExecuteScalar
Delete	CreateTable

Mapping

- Nuget package **Mapperly**

Database

DEMO

Which storage option would you use?

- Options: Preferences, Secure Storage, Database
- Examples:
 - Authentication
 - Selecting color schema
 - Saving a recipe
 - Saving settings about how items are ordered in a list (alphabetically, by date created...)

Converters

- Convert value of 1 type to another type
- Interface **IValueConverter**
- CommunityToolkit.Mvvm - **BaseConverterOneWay**
- **IMultiValueConverter**

Converters

DEMO